



Department of Telecommunications
Telecommunication Engineering Centre
K. L. Bhawan, Janpath, New Delhi – 110001
(Convergence & Broadcasting Division)



No: 4-16/2018-C&B/TEC/NWG-9

Dated: 06 April 2023

Sub: Minutes of the Seventh Meeting of National Working Group-9 (NWG-9) regarding "Broadband Cable and TV" held on 29.03.2023.

The seventh meeting of the National Working Group-9 (NWG-9) corresponding to the ITU- T Study Group 9 (SG9) constituted for the Study Period 2022-24 regarding "Broadband Cable and TV" was held on 29-03-2022 under the chairmanship of Sh. Avinash Agarwal, DDG (Convergence & Broadcasting), TEC through video-conferencing. List of participants is placed at Annexure-I.

2. At the outset, Sh. Harsh Sharma, Director (C&B-II) TEC & Convener of NWG-9, welcomed all the participants and briefly explained about the previous meetings held in Nov-2022 and agenda of the meeting.

3. Shri Avinash Agarwal, Chairman NWG-9 & DDG (C&B) TEC, in his opening remarks, stated that the next meeting of ITU-T Study Group 9 (SG-9) (Audiovisual content transmission and integrated broadband cable networks) is being held in India at IISc Bangalore from 9-18 May 2023. TEC/DoT and IISc Bangalore are jointly hosting this meeting. Accordingly, he emphasized that the contributions from NWG-9 members in ITU-T need to be enhanced. He said that this is the right time and opportunity for members to identify the work items that are relevant for India, and work upon submitting more contributions in this ITU-T SG9 meeting. He also conveyed that recently as per the direction of DoT HQ, TEC has initiated the work of monitoring/tracking the Indian patents in telecom sector as part of pre-standardization activities. It includes tracking/ monitoring of patents in the telecom sector published by/ granted to Indian companies, organisations, academia, PSUs, TSPs etc. in the emergent technologies.

4. Prof. Dr. Pradipta Biswas, IISc Bengaluru & Vice-Chair SG9/NWG-9, conveyed briefly about the Study Group-9 and hosting of its next meeting at IISc Bangalore from 9-18 May 2023. He conveyed that ITU-T has already issued the date and question-wise details of the whole programme vide TSB Collective Letter dated 03.03.2023. All members were requested to register for this SG-9 meeting. He also conveyed the structure of SG9 comprising 11 Questions, which categorised into three Working Parties. SG9 meeting will start with Opening Plenary and end with Closing Plenary. During these 10 days, Question-wise meetings and Working Parties Plenary sessions will also be conducted.

Further, he said that a workshop on the “Future of TV for South Asia, Arab and Africa” is also scheduled on **11 May 2023** from 0930 to 1800 hours. The aim of the workshop is to discuss the future of television for South Asia, Arab and Africa regions with relevant stakeholders, covering regulatory and policy frameworks, emerging and convergent ICT Infrastructures and services, as well as user interfaces and human factors issues.

Dr. Biswas conveyed regarding planning of an exhibition during this SG-9 meeting to demonstrate various telecom products/services, reports, use case etc. He also conveyed that after SG-9 meeting, IISc and DoT/TEC has planned to organise a 6G Workshop in the month of July-2023. He also invited members for participation of this workshop.

5. Further, open house discussion (as mentioned in agenda) was started. Shri Avinash Agarwal, Chairman NWG-9 & DDG (C&B) TEC conveyed that no new contribution was received for this meeting. He expressed concerns that many members are not attending the NWG9 meetings regularly. He also requested all the members to identify relevant experts and nominate subject experts/ academia/practitioners as members of NWG9. He conveyed about the mode of SG-9 meeting, which is going to be held in physical and remotely. He requested to NWG members to register themselves for this meeting. The Link of registration is as below;

<https://www.itu.int/net4/CRM/xreg/web/Login.aspx?src=Registration&Event=C-00012396>

6. Shri Avinash Agarwal conveyed that any person (including non NWG members) who is interested in SG-9 topics may attend the workshop scheduled on 11.05.2023. There is no need to become a member of NWG for this purpose. He also shared that perhaps Hon’ble MoC, Hon’ble MoSC and some other distinguished top official may be the invited guests for this workshop. The workshop would be attended by national and global experts from **South Asia, Arab and Africa Regions** from industries, academia, R&D centres, start-ups and government to share recent developments in this field. The workshop will be in both online and offline mode.

7. On potential contribution/ongoing work items, Shri Avinash Agarwal requested that respective members/contributors should attend the meeting in physical mode. He told that all contribution should be prepared and submitted to the Convener/ Chairman NWG9 before the next meeting of NWG9, which may be scheduled in 2nd week of April 2023. Following are the ongoing work items that were initiated by India in the past meetings:

Sno.	Question/Topic name
1.	Q1/9 <i>‘J.TR.WiFiTV (Technical Report) on Secondary distribution of digital television and audiovisual content to portable devices using Wi-Fi’</i>
2.	Q2/9 <i>TR-FSR (Technical report) regarding ‘Factual subscriber base reporting and protected content delivery in conditional access system’</i>

3.	Q2/9 J.FSR-REQ regarding <i>'Factual subscriber-base reporting and protected content delivery in conditional access system – Requirements'</i>
4.	Q11/9 <i>JSTR.LCAP (Technical Report) Technical advances, challenges, and best practices in live captioning</i>
5.	Q 11/9 <i>Contribution on "Common user profile format for audio-visual content distribution"</i>

8. Sh. Avinash Agarwal emphasized to identify some new work items/topics for preparation of contributions. He requested to all NWG-9 members at various management positions in ITU-T SG-9 (Vice-chairman, Rapporteur, Associate Rapporteur, Editors etc. and academia to guide in this matter.

9. Prof. J. S. Sahambi, IIT Ropar submitted that industry and academia should come forward for standardization activities and prepare the contribution along with govt. administration. On request of chair, he agreed to co-ordinate with other faculty members, other IITs, research institute and compile the data regarding the research work (including patents) which have been done or going on in the field of convergence and broadcasting. He will share the data to NWG-9 for further process.

10. Further, all participants interacted one by one. Dr. Shiv, Broadband India Forum (BIF) submitted that he will do some work on different questions and will share some meaningful contribution with NWG before its next meeting.

11. Ms. Meenakshi Singhvi, Director, AIR & Doordarshan, Prasar Bharti Mumbai conveyed that she is working on a contribution to Q6/9 (Functional requirements for terminal devices of the integrated broadband cable network) and has contacted some agencies including small companies which are manufacturing te setup boxes, etc. Few of them have already submitted the feedback to improve the ITU-T Recommendation on Hybrid Setup Box and what changes are required in terms of compatibility and futuristic demand. Based on the feedback, she will prepare the contribution and share with NWG-9. She also invited other members/volunteers to work on this contribution.

Sh. Aviansh Agarwal suggested that representatives of such manufacturing companies working in this field may also be invited to join the NWG-9. Ms. Meenakshi Singhvi agreed to share the details of such companies.

12. Sh. Pallabh Dutta, C-DoT conveyed that C-DoT will surely participate in the May 2023 SG-9 meeting and the Workshop scheduled on 11.05.2023 and will exhibit/demo

some use cases. Sh. Avinash Agarwal suggested that C-DoT may show following two use cases (related to Indian contributions in SG9) in this exhibition;

- (a) Converged Access Node for Broadcast through WiFi
- (b) CAS and SMS devices

Sh. Pallabh agreed on this and submitted that contribution regarding converged access node for broadcasting through WiFi will be prepared in consultation with TEC. Recently TEC has also released a GR documents on this topic.

13. Sh. Avinash Agarwal introduced one new member Dr. Prem Chand Jain to all NWG members. Sh. Prem Chand Jain is having more than 40 years experience in the field of Convergence and Broadcasting including the development of setup box in India. He agreed to contribute in this NWG-9. Sh. Avinash Agarwal requested him for guiding the NWG-9 with his vast experience for meeting the Indian requirements.

14. Shri Avinash Agarwal expressed concerns that many members are not attending the NWG9 meetings regularly. He also requested all the members to give some idea or mechanism to improve the active participation. On this concerns, Prof. J. S. Sahambi, IIT Ropar agreed to share some idea with NWG9. Sh. Rajiv Khattar suggested that like BIS, we may circulate the attendance sheet with previous meeting attendance and also fix minimum attending criterion (3 meetings) for this NWG-9 and if someone is not fulfilling this then the same may be removed from NWG-9.

15. Sh. Rajiv Khattar submitted that he is working on Q2/9 TR-FSR (Technical report) regarding '*Factual subscriber base reporting and protected content delivery in conditional access system*' and will finalize shortly within next 7 days. He also agreed to check the possibility of new work items related to the testing of CAS and SMS system.

16. Sh. Sundar Narayan referred the Q3/9 – AI enabled enhanced functions over integrated broadband cable network and agreed to submit a brief note regarding a new work item.

17. Sh. Sunil Gautam, Director (SPPI), DoT HQ submitted that he will check and will submit some details regarding lawful interception and AI before the next meeting.

18. Sh. Avinash Agarwal asked to give the details of patents related to the SG-9 topics/ fields from CDoT, IISc, and other members. In addition, he also highlighted further developments in the Standards Coordination Portal of TEC. He stated that in next few days information will be conveyed to all NWG members regarding registration and how to use this portal. SCP portal will be very useful and all ITU-T information can be downloaded even if someone does not have TIES account, which is required for accessing the documents /information from ITU-T website. Standards Coordination portal of TEC will be a good repository of work for Indian contributions.

19. Sh. Avinash Agarwal asked to participants to volunteer for organizing the SG9 meeting in Bangalore. In response, Sh, Pallabh Dutta, CDoT, Sh. Sunil, Sh. Sundar Narayan and Dr. Shiv, BIF agreed to contribute as volunteers in the upcoming meeting of SG-9.

20. In last, the meeting ended with the vote of thanks to the chair.



(Harsh Sharma)

Convener, NWG-9 & Director, (C&B-II), TEC

Encl.: Annexure-I

To,

All members of the NWG-9 and all the Participants.

Copy for kind information to:

1. Sr. DDG & Head TEC, New Delhi
2. Chairman NWG-9 & DDG (C&B), TEC, New Delhi.

ANNEXURE-I

List of Participants:

1. Sh. Avinash Agarwal, Chairman NWG-9 & DDG (C&B), TEC.
2. Dr. Pradipta Biswas, IISc Bengaluru & Vice-Chair NWG-9.
3. Ms. Meenakshi Singhvi, Director, AIR & Doordarshan, Prasar Bharti Mumbai
4. Sh. Rajiv Khattar, Researcher & Industry specialist
5. Prof. J. S. Sahambi, IIT Ropar
6. Sh. Pallabh Dutta, C-DoT
7. Sh. Sundar Narayanan, Researcher & Consultant
8. Dr. Shiv, Broadband India Forum (BIF)
9. Dr. Prem Chand Jain, Professor(Guest), Shiv Nadar University
10. Sh. M Rajendran
11. Sh. Rakesh Desai, DDG(FA), TEC
12. Sh. Sunil Gautam, Director (SPPI), DoT HQ
13. Sh. Aftab Alam, Director USOF, DoT HQ
14. Sh. Naveen Khichar, ADET, DoT HQ
15. Sh. Brajesh Kumar, Director (C&B)-I, TEC.
16. Sh. Harsh Sharma, Convener, NWG-9 & Director (C&B)-II, TEC.
